Prediction Market Synthetics

Andrew, Drishti, Clinton, Derek, Anirudh, Yiming

Hypotheses

1] Let Sw and Sl be the conditional probabilities that an arbitrary team A win the series conditioned on winning / losing the current game respectively. Market makers may not be pricing this conditional probability accurately.

Can we trade this using synthetics?

2] Given current game data, can we market make more intelligently on the series market, while hedging out any delta exposure to the game market itself.

Can we trade *this* using synthetics?

Data

Kalshi

- 30,931 NBA games, 2002-03 to 2025-26, ESPN
- NBA Series, 360 playoffs series
- NHL (27,201 games, 2005-06 to 2025-26)
- NHL Series, 491 playoffs series
- NBA Official Injury Report (2022-2026)
  - first report at or after 17:30 ET the day before the game
  - latest report at or before tip minus 60 minutes
- Player Box Scores (467,380 player-games, 2012-13 to 2025-26)



Polymarket

Data

- Kalshi:
  - 768 contracts - 2025, 2026, NBA + NHL Playoffs Games + Series
  - 2.05 M rows (minute level granularity - candlesticks)
  - 2.03 M rows (trades made)
- Polymarket:
  - 85 games + 15 series (2026 NBA), 82 games + 15 series (2026 NHL)
  - Second-by-second order book data (free API)
  - NBA Trades (686k rows)

What is a Synthetic?

Multiple positions that cancel out over certain intervals to create exposure to certain market dynamics.

- Call Option + Short Stock
- Put Option + Long Stock

Create the payoff of a financial instrument using other financial instruments.

Consistency Formulas

Fair series price = g * Sw + (1 - g) * Sl

Sw = P(A wins series | A wins this game)
Sl = P(A wins series | A loses this game)

F = Sl + g * [Sw - Sl]
F = Sw - (1 - g) * [Sw - Sl]

Assuming we fix Sw and Sl:

How do we construct a synthetic to hedge out the game outcome?

Taker Baseline (hypothesis #1)

We start by creating a naive baseline for testing our first hypothesis that market makers fail to set the fair value of conditional outcomes…

- Let *g* denote the probability of a given team winning the first game.

- Approach A: We estimate the probability of that team winning the series using only *g* as the implied probability of winning each game.

- Approach B: We follow the same strategy as before, except adjust *g* between games and use that updated *g* to predict the odds of winning the series.

P&L of Trading Mid-Game vs. Only Between

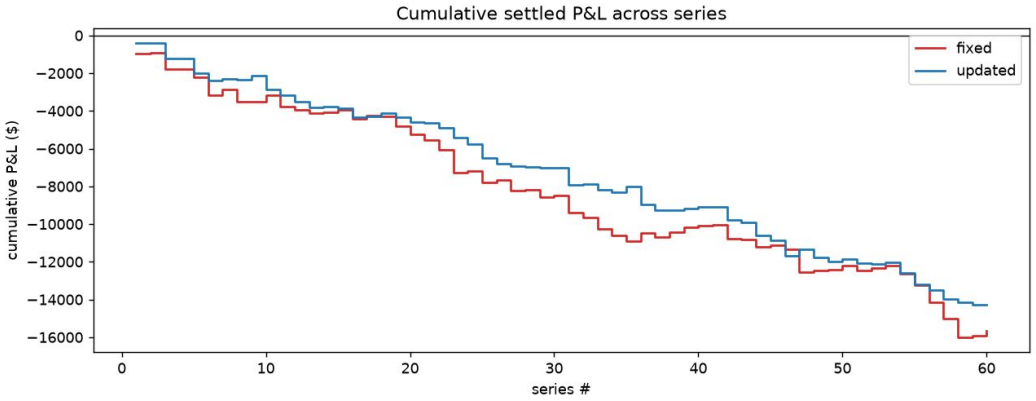
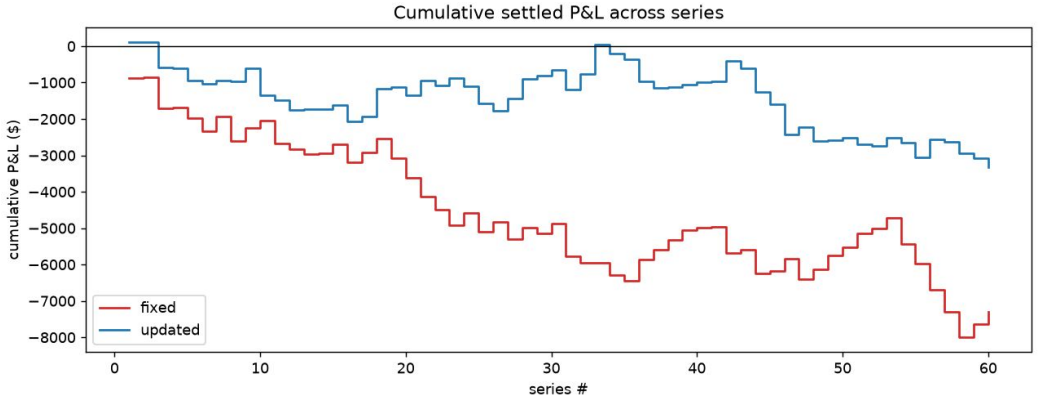


Fig. 1: Baseline trading both during games and between games on Kalshi.

Fig. 2: Baseline trading only between games with on Kalshi.



P&L of Trading NBA vs. NHL (Polymarket)

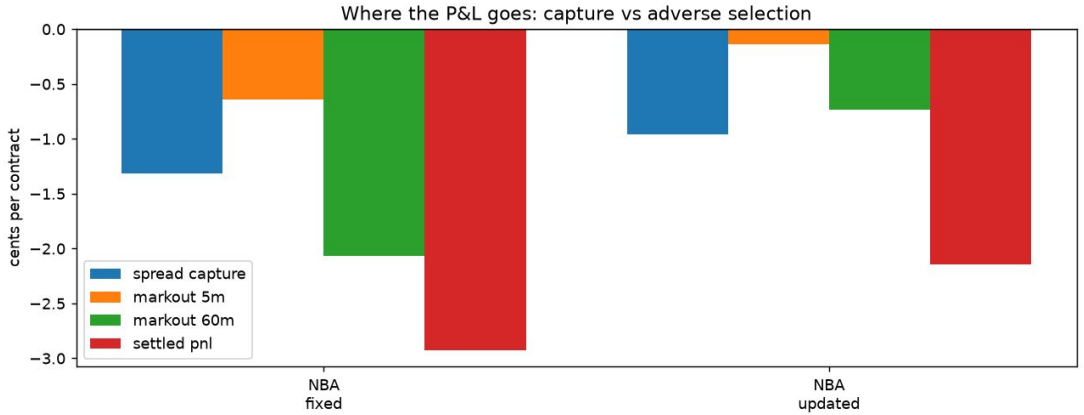
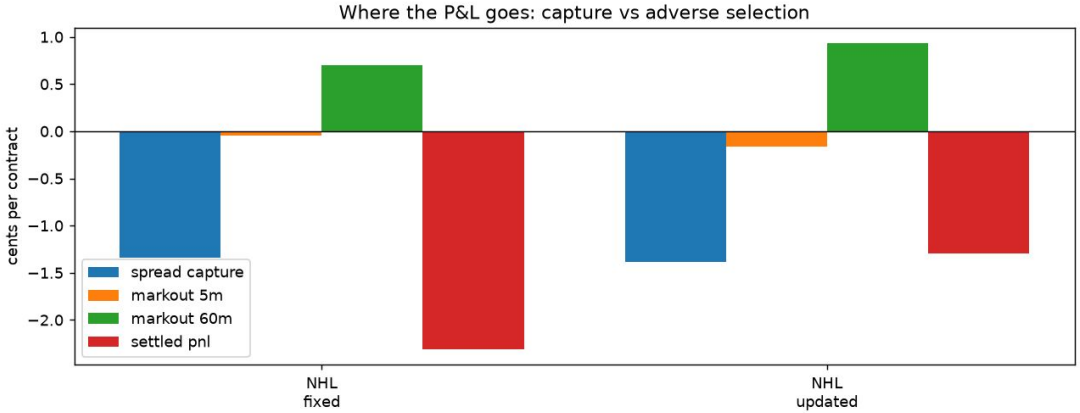
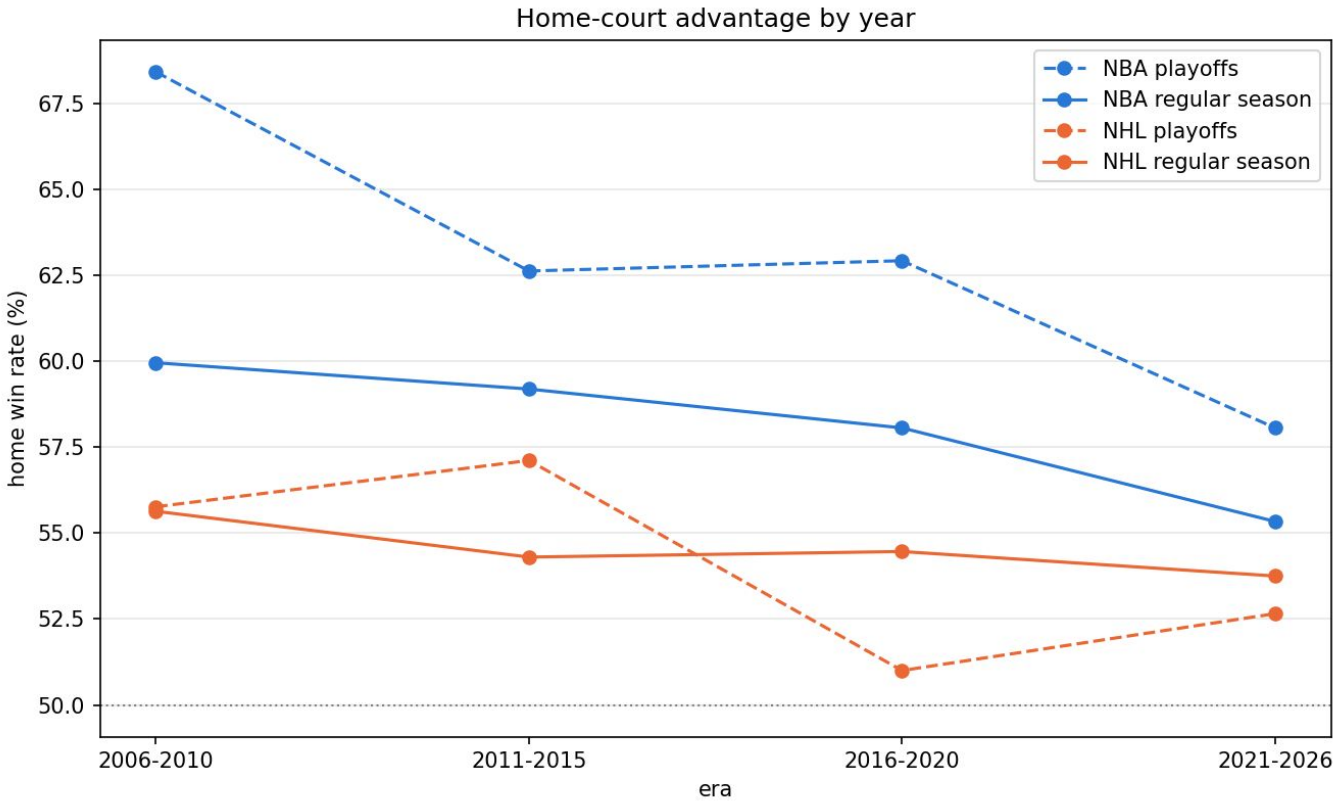


Fig. 4: Details of baseline strategies on NBA data.

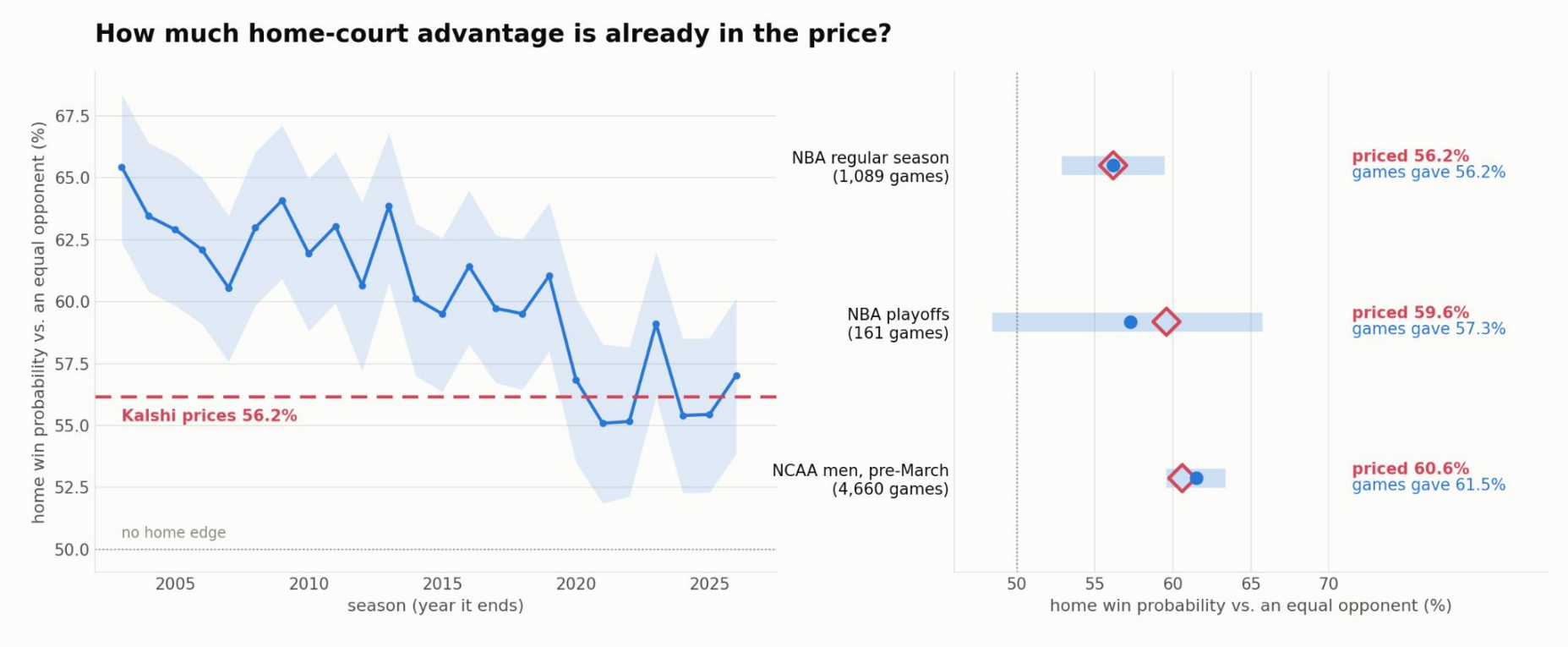
Fig. 3: Details of baseline strategies on NHL data.



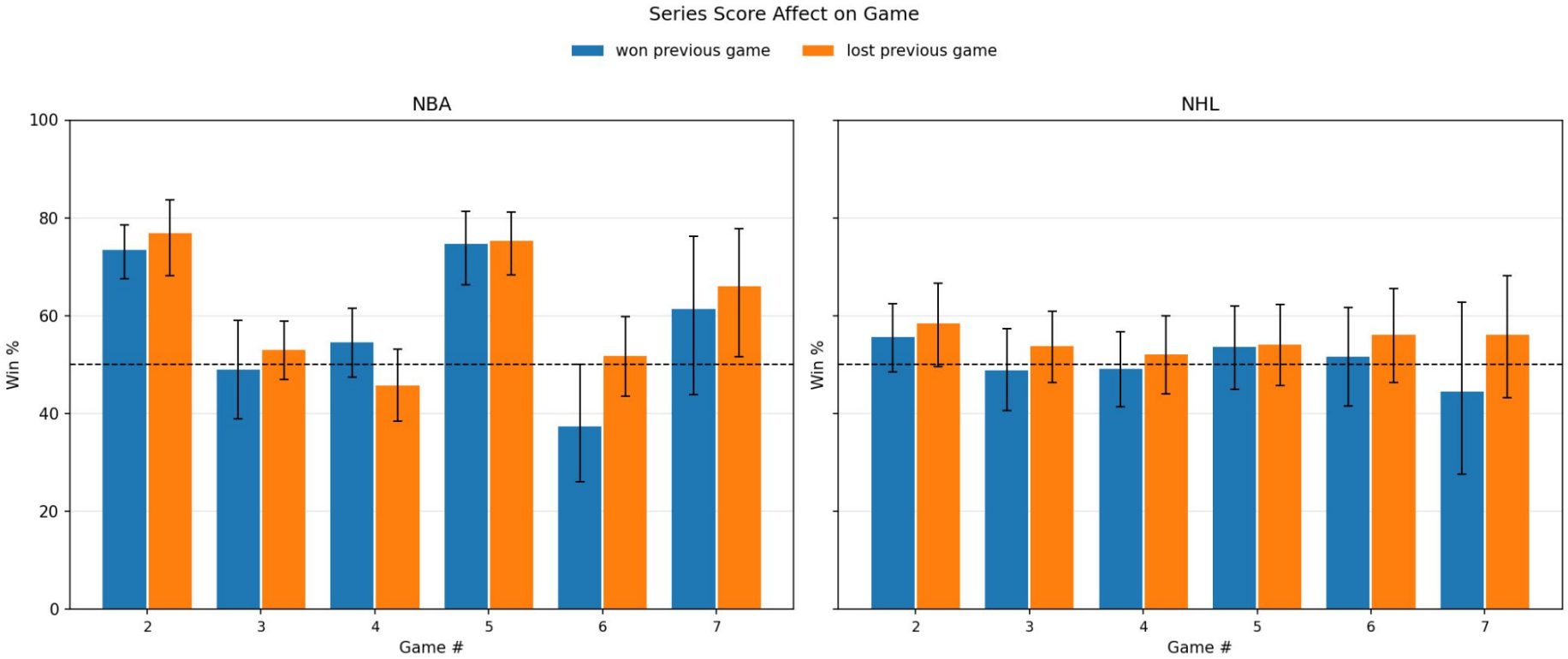
More Features



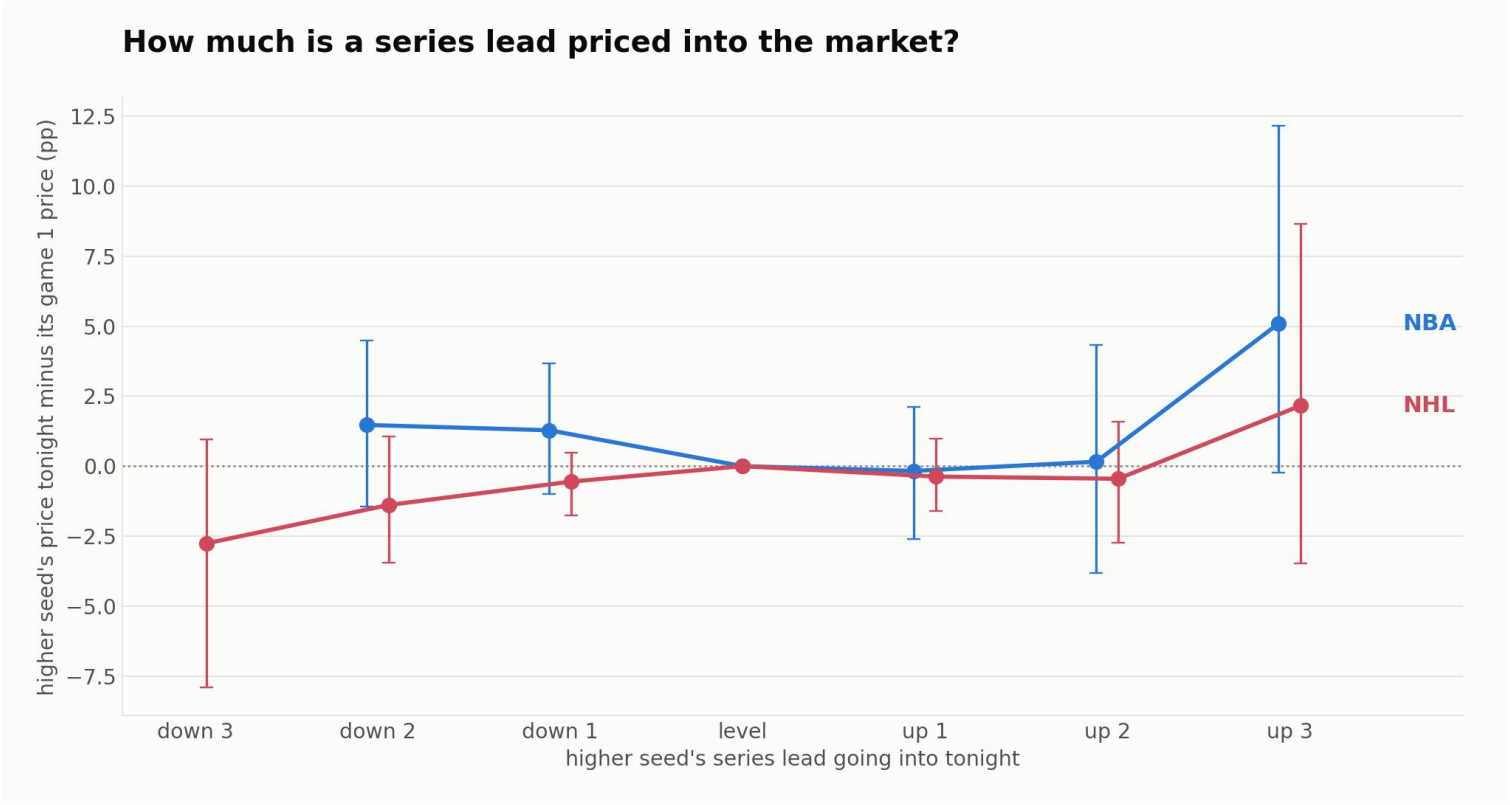
Market Pricing of Homecourt (Derek)



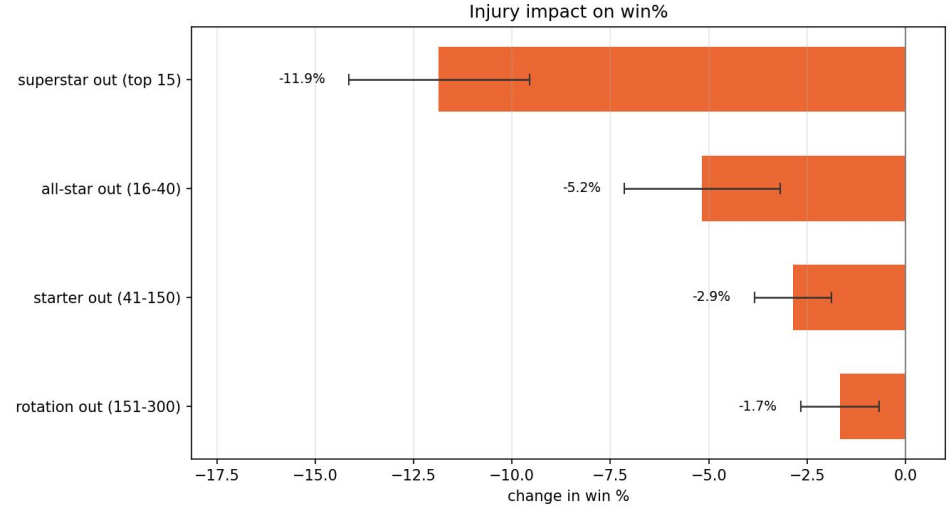
Series Effect

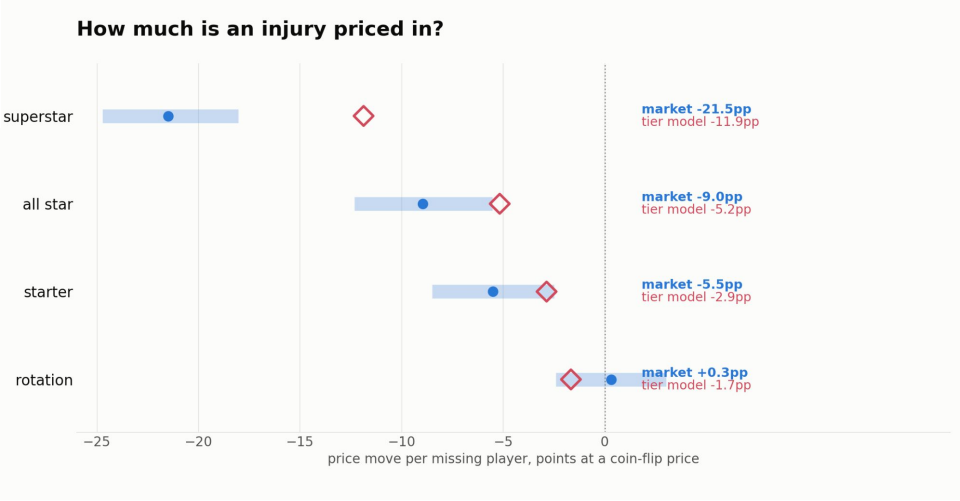


Series Effect

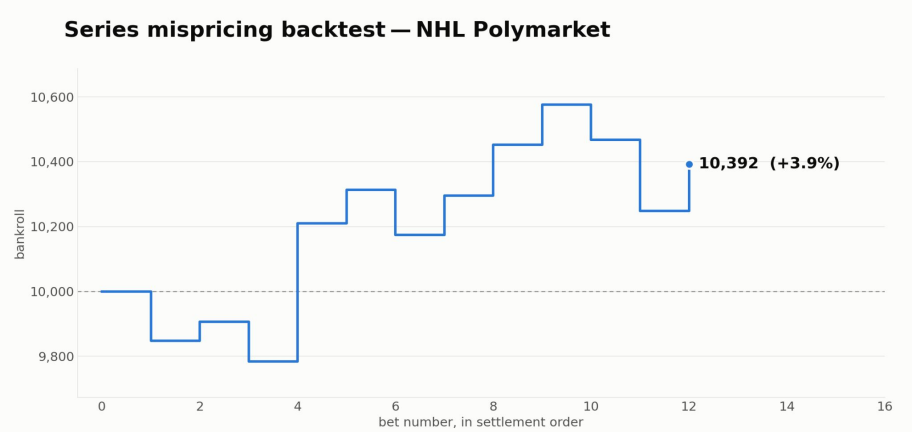


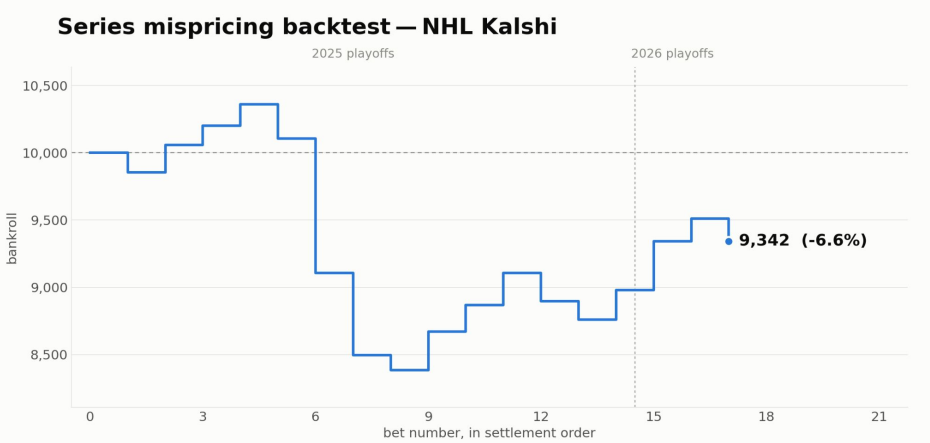
Injuries

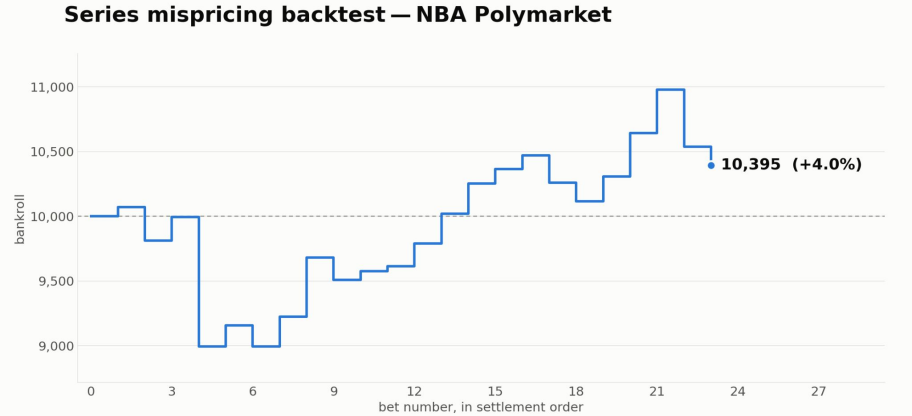

Injury impact on win%
superstar out (top 15)
all-star out (16-40)
starter out (41-150)
rotation out (151-300)
-11.9%
-5.2%
-2.9%
-1.7%
−17.5
−15.0
−12.5
−10.0
−7.5
−5.0
−2.5
0.0
change in win %

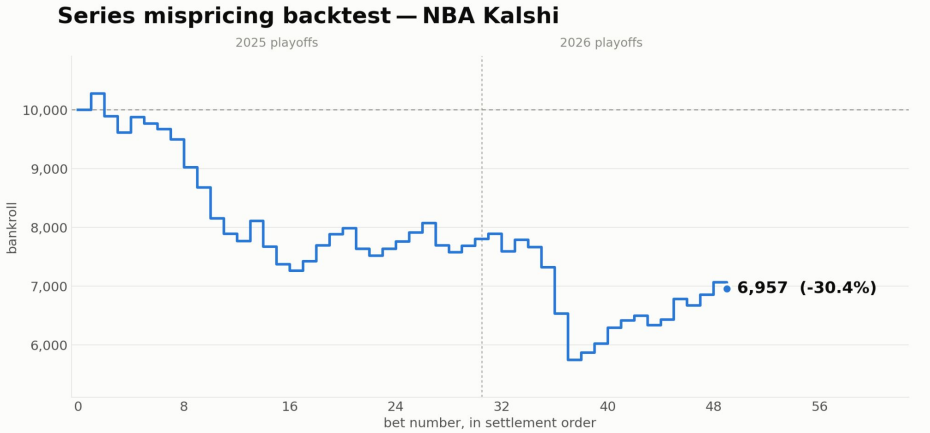

How much is an injury priced in?
superstar
all star
starter
rotation
market -21.5pp
tier model -11.9pp
market -9.0pp
tier model -5.2pp
market -5.5pp
tier model -2.9pp
market +0.3pp
tier model -1.7pp
−25
−20
−15
−10
−5
0
price move per missing player, points at a coin-flip price

Series Mispricing









Series Mispricing (Derek)

Market	Trades	Volume	P&L	P&L per trade	Drawdown	Profit
NBA Kalshi	49	16,820	−3,043	−62	−41.4%	−30.4%
NBA Polymarket	23	11,499	+395	+17	−10.7%	+4.0%
NHL Kalshi	17	6,308	−658	−39	−17.8%	−6.6%
NHL Polymarket	12	4,409	+392	+33	−3.1%	+3.9%
All pooled	100	29,855	−3,406	−34	−54.8%	−34.1%

Basketball Model (Anirudh + Yiming)

Q: Can we train a model to predict game and series outcomes?

Predictions:

- Per-Game prediction
- Mid-Series result prediction

Approaches

- Logistic Regression
- GBM (Gradient Boosting)

Baseline Comparison (Naive Approach)

Elo Comparison - Higher Elo = Win

How did it do (Combined for Kalshi + Polymarket)

Log Loss of Naive Approach (Series): 0.652

Log Loss of Market (Series): 0.576

Log Loss of Naive Approach (Game): 0.667

Log Loss of Market (Game): 0.663

Input:

Difference in:

elo, net rating, win rate, prev game scoring, pace, rest days, b2b status, games played recently, true shooting %, turnover rate, offensive-rebound rate, free-throw rate

As well as court data:

Altitude, home-court, playoff-game, series lead, and series margin.

Game Model

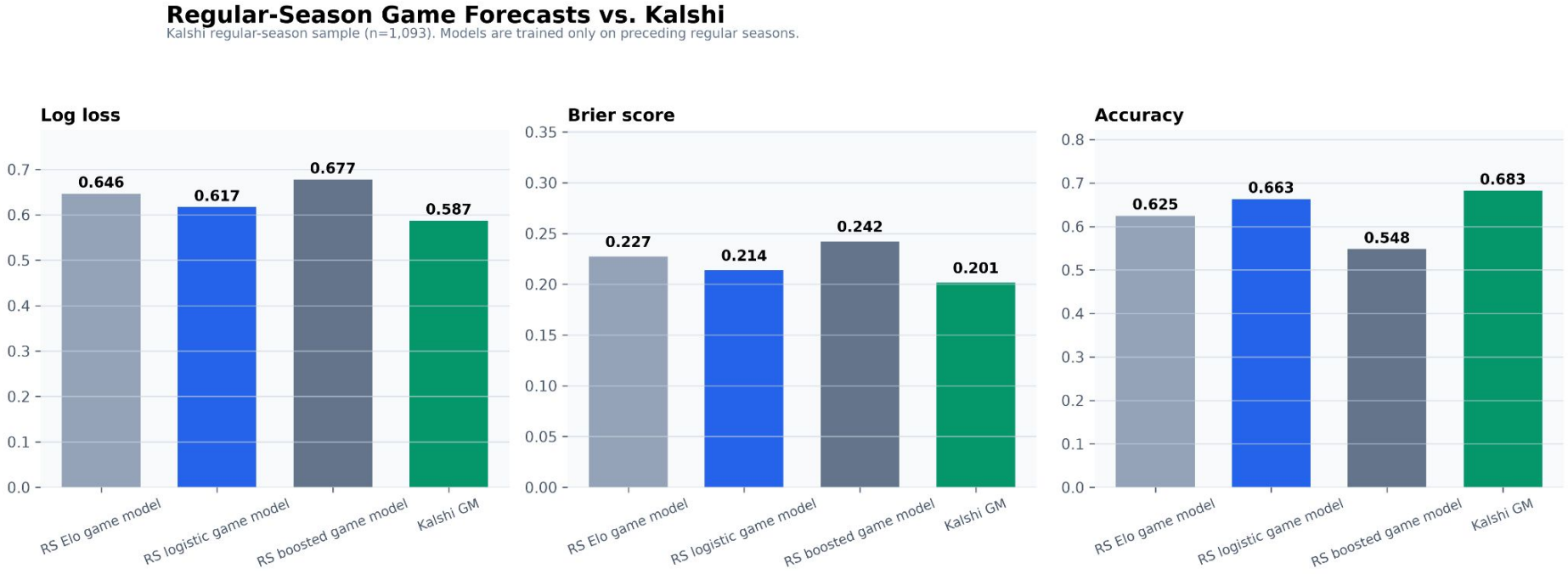
- Target: Predict g
- Outcome of the current game
- Sw and Sl given
- Later plugged into series formula to find probability of winning a series

$$\text{Fair series price} = g * Sw + (1 - g) * Sl$$

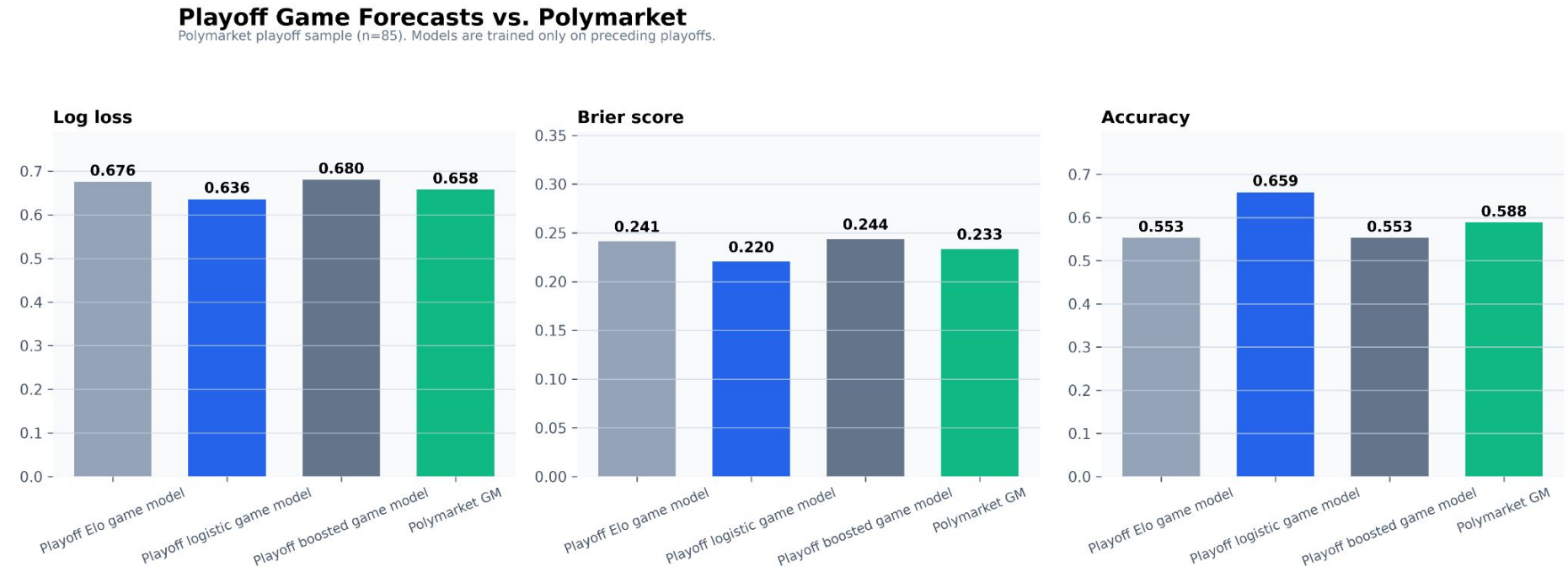
$$Sw = P(\text{A wins series} \mid \text{A wins this game})$$
$$Sl = P(\text{A wins series} \mid \text{A loses this game})$$

$$F = Sl + g * [Sw - Sl]$$
$$F = Sw - (1 - g) * [Sw - Sl]$$

Game Model Results

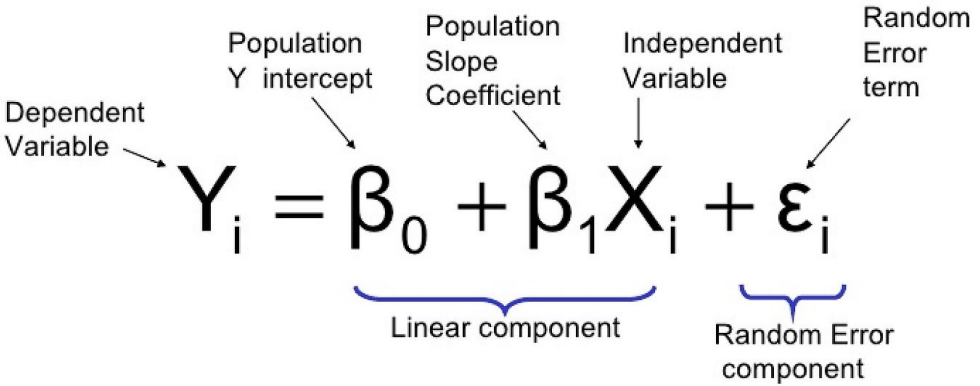


Game Model Results

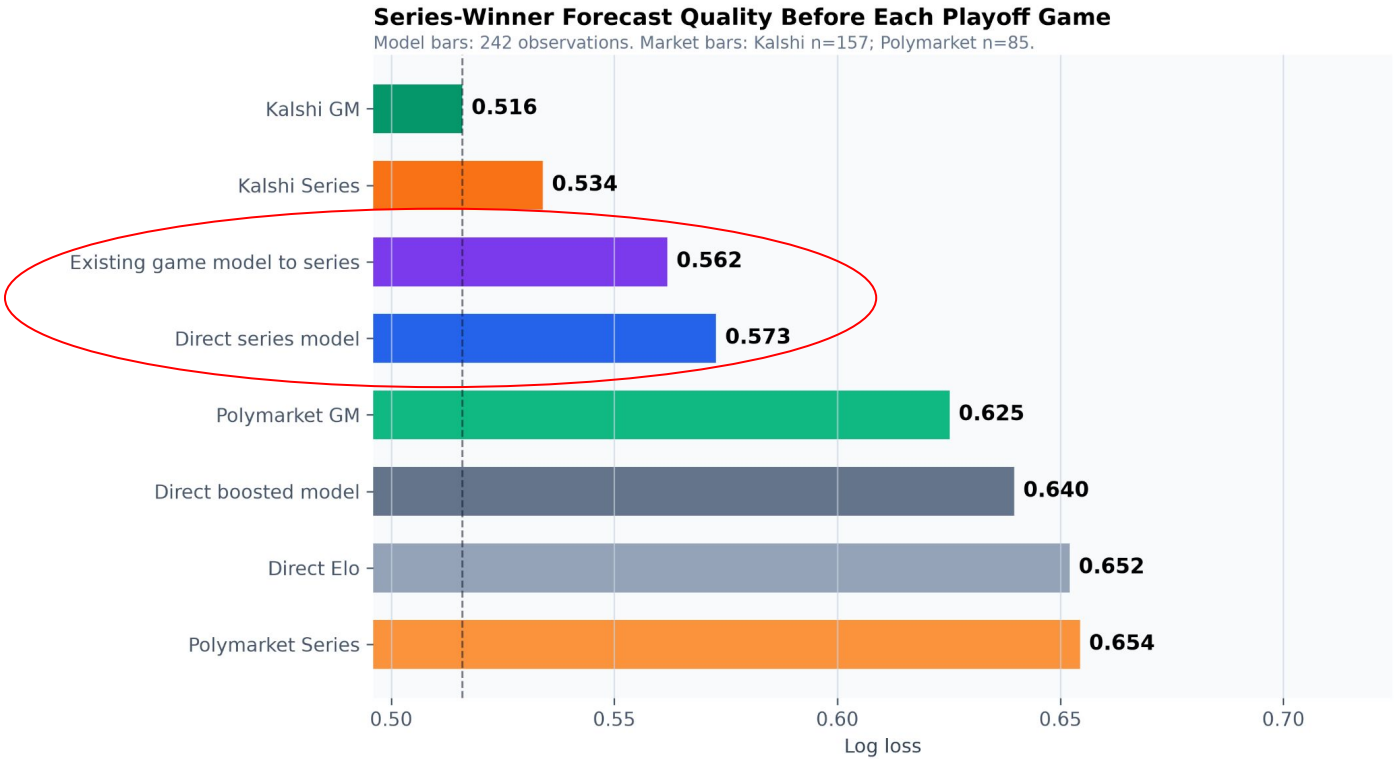


Series Model

- Target: Find series winner
- Didn’t try to predict game winner
- Logistic regression, tree-boosted, Elo only methods



Series Model Results




Where the Direct and Recursive Forecasts Disagree
One point per underlying game (n=161).
Team A lost series
Team A won series
Equal forecast
Direct series-model probability
Existing game-model to series probability
0.0
0.2
0.4
0.6
0.8
1.0

Model Metrics

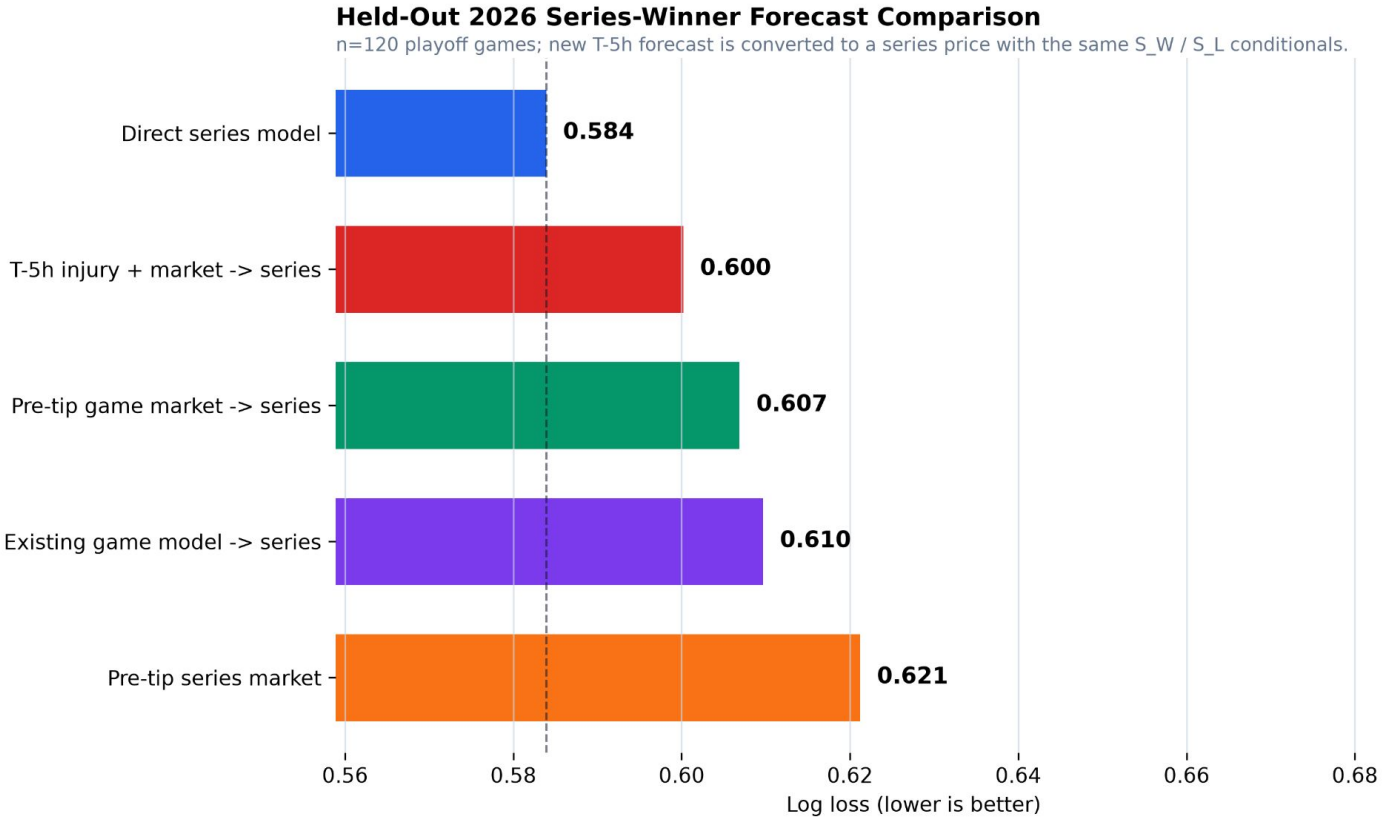
Game

Model input	Weight per 1 SD
True shooting difference	+0.245
Elo difference	+0.128
Playoff Elo interaction	+0.128
Adjusted net-rating difference	+0.127
Home-court context	+0.113
Opponent offensive-rebound difference	-0.075
Opponent true-shooting difference	+0.064
Opponent turnover-rate difference	+0.052

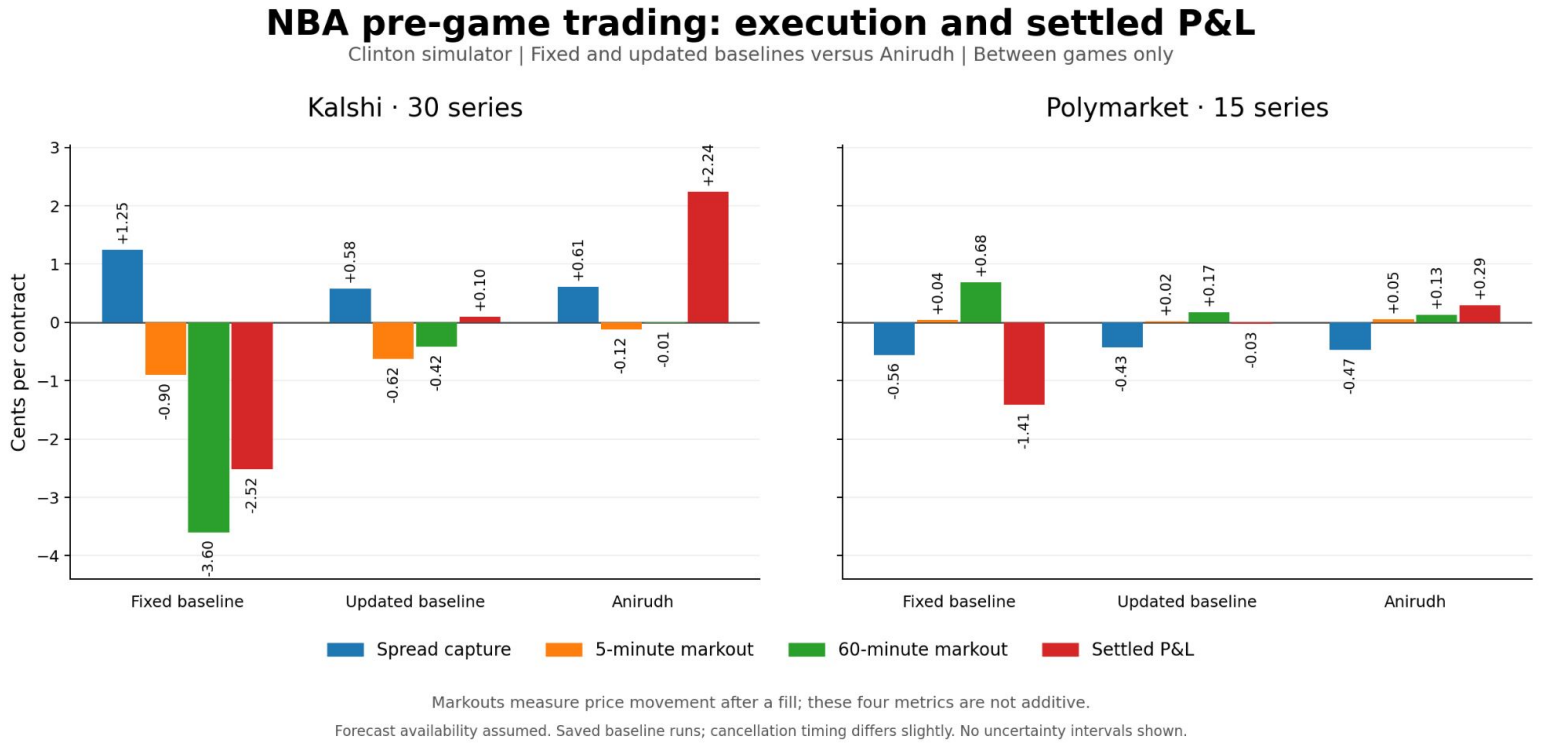
Series

Model input	Weight per 1 SD
True shooting difference	+0.239
Current series lead	+0.225
Elo difference	+0.201
Away team's series wins	-0.176
Net-rating difference	+0.162
Free-throw-rate difference	-0.155
Adjusted net-rating difference	+0.151
Last-10 win-rate difference	+0.134

With Injury Data



Trading on the Basketball Model (Anirudh + Yiming)



Markouts measure price movement after a fill; these four metrics are not additive.

Forecast availability assumed. Saved baseline runs; cancellation timing differs slightly. No uncertainty intervals shown.

~~F~~Maker Baseline (hypothesis #~~1~~2)

We start by ~~creating~~ continuing to use a naive baseline for testing our ~~first~~ second hypothesis that ~~market makers fail to set the fair value of conditional outcomes~~ actual game data can help more intelligently make a market…

- We extend each of Approach A and Approach B, continuing to use the market-priced g to find an implied probability of winning each game, either adjusted or unadjusted.

- Now, instead of just using a taker strategy at fair value, we make a market at a half-spread of 1%.

P&L of Bounded vs. Unbounded Clip Size

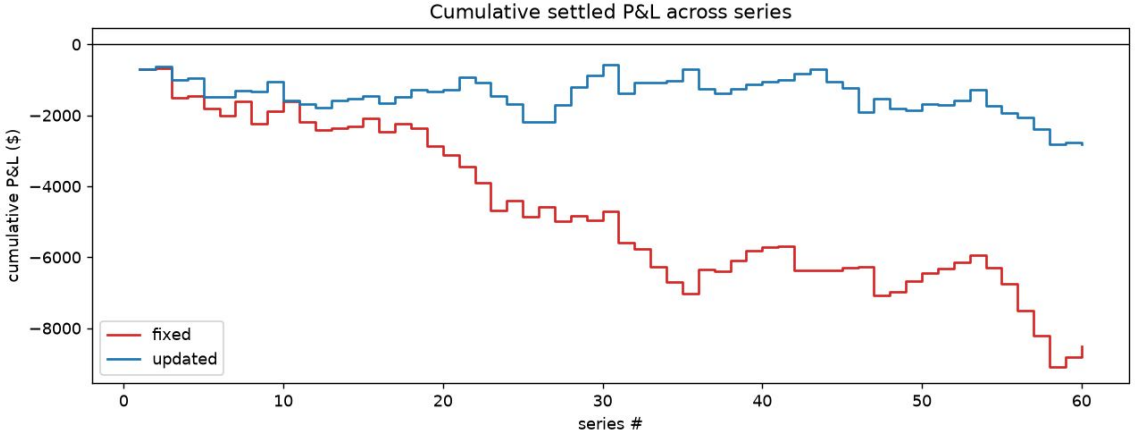
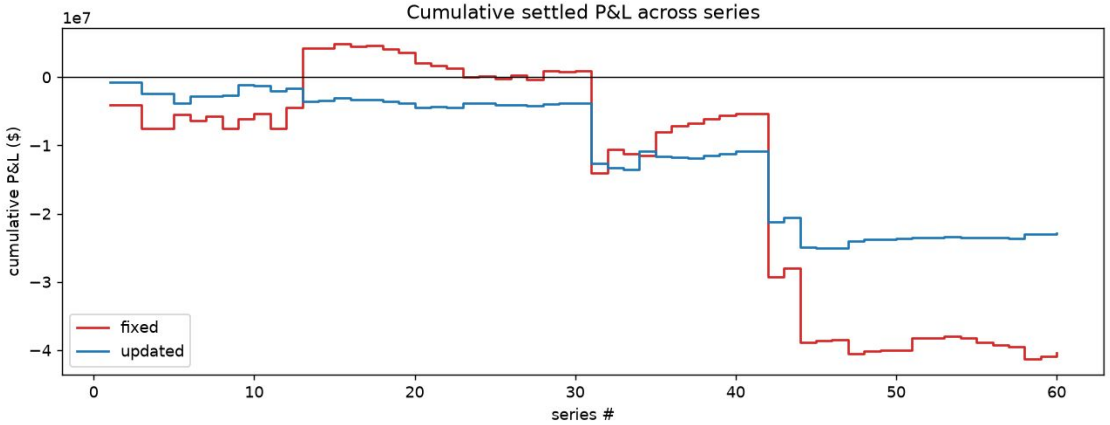


Fig. 5: Using clip size of 100 and a maximum position (ceiling on inventory) of 1000 on Kalshi.

Fig. 6: Trades unlimited volumes of each available contract on Kalshi



P&L of 0.01 vs. 0.04 Spread (Polymarket)

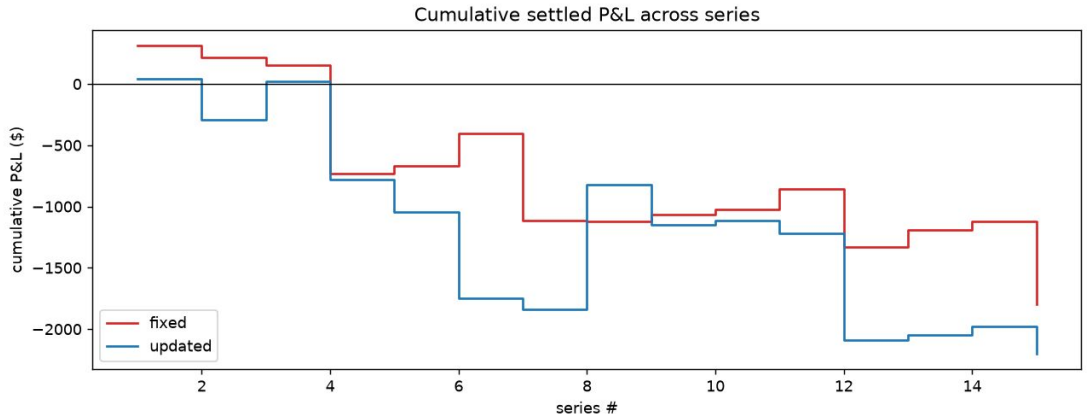


Fig. 7: 0.005 half-spread on NBA data.

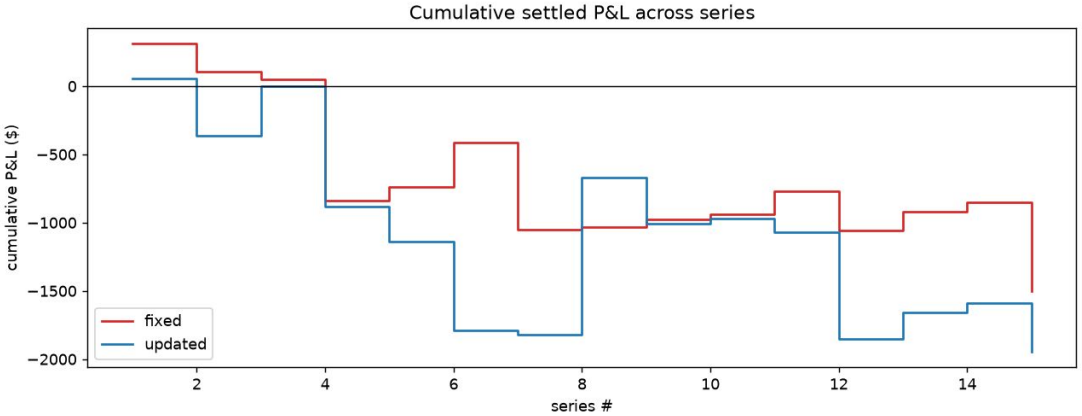


Fig. 8: 0.02 half-spread on NBA data.

Market Making Series Delta Approach

series_delta(t) = alpha * game_delta(t)

game_delta(t) → exponential weighted avg of last j frames

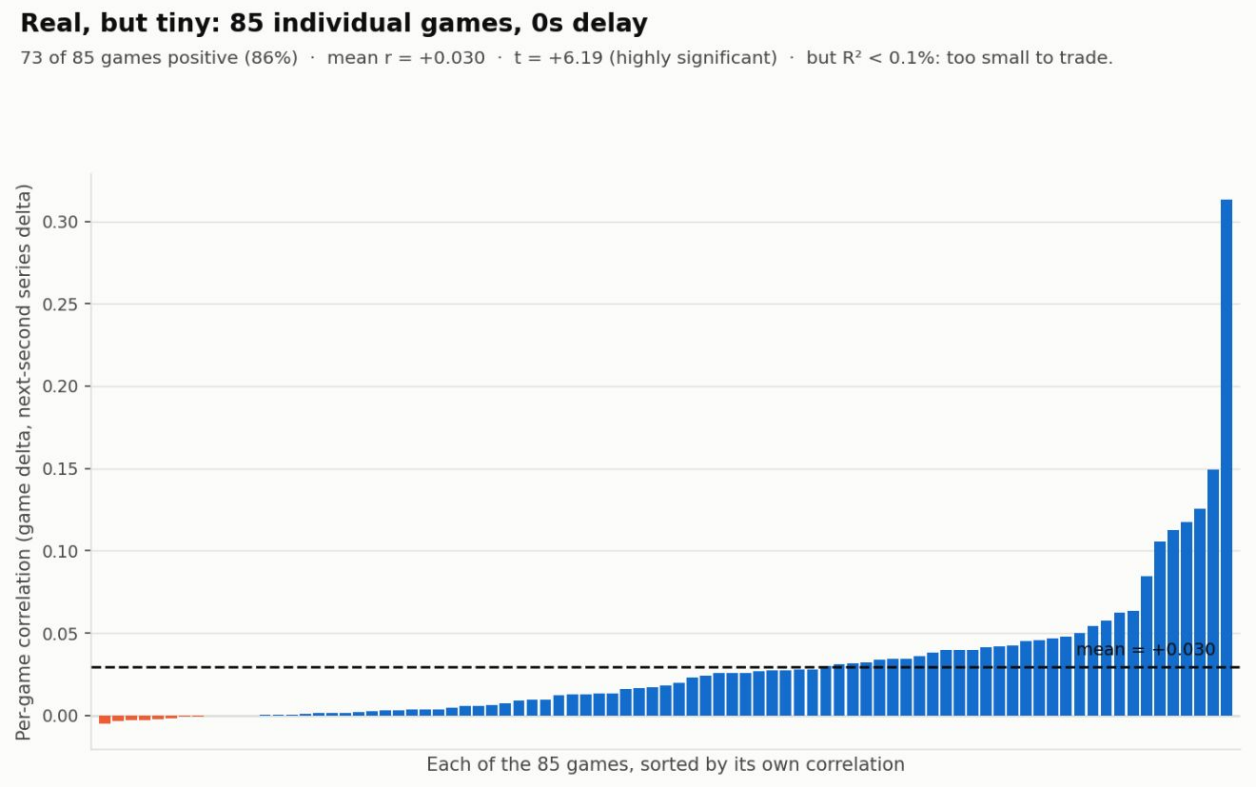
- Frame 1s before * B
- Frame 2s before * B^2
- …
- Frame js before * B^j

} Divide sum by B + B^2 .. + B^j

Tuning

- Alpha: 0.03, Beta: 0.9, j = 3.

Market Making



Comparing Approaches

